

**a** Per-operator AUROC (task-split mean)

|                       |     |      |       |        |        |          |        |       |           |       |
|-----------------------|-----|------|-------|--------|--------|----------|--------|-------|-----------|-------|
| insole missing [S]    | .88 | .89  | .47   | .45    | .88    | .86      | .42    | .85   | .50       | .31   |
| encoder dropout [S]   | .92 | .92  | .62   | .44    | .77    | .75      | .49    | .96   | .50       | .49   |
| IMU bias [S]          | .50 | .50  | .53   | .46    | .51    | .51      | .48    | .50   | .50       | .51   |
| packet loss [S]       | .83 | .85  | .50   | .50    | .50    | .51      | .49    | .95   | .47       | .50   |
| stuck IMU [S]         | .90 | .90  | .48   | .53    | .57    | .56      | .52    | .98   | 1.00      | .50   |
| sensor delay [S]      | .83 | .59  | .46   | .54    | .58    | .57      | .52    | .50   | .97       | .51   |
| burst packet loss [H] | .75 | .83  | .50   | .50    | .50    | .51      | .50    | .92   | .05       | .50   |
| partial dropout [H]   | .86 | .86  | .50   | .50    | .50    | .51      | .50    | .95   | 1.00      | .50   |
| delay jitter [H]      | .82 | .60  | .46   | .54    | .58    | .57      | .51    | .50   | .95       | .51   |
|                       | All | Gate | Fault | Aleat. | Recon. | Forecast | Epist. | Stale | Coherence | Drift |

**b** Macro AUROC